

Why Did Bonds Stop Hedging Stocks?

Evidence on the Post-2021 Shift in Stock–Bond Comovement

Anonymized for Review

Abstract

For much of 2003 to 2020, long-duration US Treasuries helped hedge equity risk, but that role weakened sharply after 2021 as the average monthly stock-bond correlation rose from -0.344 to $+0.033$. The standard explanation emphasizes inflation and the vulnerability of nominal bonds. This paper asks whether that explanation is sufficient, or whether the equity market’s response to growth-sensitive rate news also changed. Using daily US data from 2003 through 2025, three observable rate-based shock proxies, and a broad set of equity baskets, the evidence indicates that the post-2021 shift is not simply an inflation story. Instead, the key change is that growth-sensitive real-rate news became materially worse for equities, consistent with a more hawkish perceived Federal Reserve reaction function. For allocators, long Treasuries are better viewed as conditional diversifiers rather than unconditional equity hedges.

Keywords: stock–bond correlation; Treasury hedging; real yields; inflation; monetary policy risk.

JEL classifications: G12; E43; E44.

1. Introduction

The stock–bond correlation is a central input in multi-asset allocation (Czaronis, Kritzman, and Turkington, 2021; Molenaar et al., 2024). When that correlation is negative, long-duration government bonds cushion equity drawdowns and raise the diversification value of a conventional 60/40 portfolio. When that correlation moves toward zero or turns positive, the same bond allocation delivers less protection precisely when investors need it most. Recent work in both the academic and practitioner literatures emphasizes that the stock–bond correlation is not stable and can shift across macro regimes (Campbell, Sunderam, and Viceira, 2017; Campbell, Pflueger, and Viceira, 2020; Czaronis, Kritzman, and Turkington, 2021; Molenaar et al., 2024).

This paper studies the post-2021 US breakdown of the stock–bond hedge. In the monthly realized-correlation series used here, the average stock–bond correlation is -0.344 over 2003–2020 and $+0.033$ over 2021–2025, a shift of $+0.377$. The shift is large in economic terms. A bond allocation that previously reduced total portfolio risk no longer offered the same protection after inflation surged and the Federal Reserve tightened (BLS, 2023; Board of Governors, n.d.).

A common explanation is that inflation became the dominant macro force. That view is well grounded in the literature: when inflation is salient, both stocks and nominal bonds can react negatively to inflation news, pushing their correlation upward (Campbell, Sunderam, and Viceira, 2017; Campbell, Pflueger, and Viceira, 2020; Molenaar et al., 2024). But inflation may not be the whole story. After 2021, markets also repriced the expected policy response to strong macro data. If stronger activity or tighter labor-market conditions increasingly implied a more hawkish policy path, then real-rate news could become bad news for equities even when the underlying macro signal was “good” (Boyd, Hu, and Jagannathan, 2005; Pflueger and Rinaldi, 2022).

The main result is not a dramatic new response of equities to the inflation proxy. It is a sign reversal in equities’ response to the real-rate proxy. In the pooled interaction regression, the market’s loading on the real-rate proxy changes from $+0.00185$ before 2021 to -0.00170 after 2021, for a differential effect of -0.00354 ($t = -5.18$). The analogous post-2021 interaction is smaller for the inflation proxy (-0.00152 , $t = -2.65$) and statistically negligible for the short-rate proxy (-0.00035 , $t = -0.53$). In a time-varying decomposition, about 78% of the observed correlation shift is associated with the real-rate channel and about 21% with the inflation channel. Put differently, the hedge weakened not only because bonds remained vulnerable to inflation. It also weakened because the stock market began treating growth-sensitive real-rate news as materially worse news for equities.

Two clarifications matter. First, the 10-year real-yield change is a *proxy*, not a clean structural growth shock. Real yields can move because of expected growth, expected policy, term premia, or risk appetite. The shorthand “growth channel” is used only because the post-2021 result is most naturally interpreted through growth-sensitive discount-rate news. Second, the decomposition is not a full structural variance decomposition. Because the shock proxies are not orthogonalized, the channel contributions summarize the own-channel component of the correlation shift and leave cross-shock covariance terms in the residual fit.

This paper uses a daily reduced-form design and a broad set of equity baskets to ask whether the post-2021 loss of Treasury hedging power is better described as an inflation-only story or as a change in the equity response to real-rate news. The evidence points to the second interpretation. For asset allocators, that shifts attention from inflation alone to the interaction between macro news and the perceived policy reaction function.

The rest of the paper is organized as follows. Section 2 describes the data and empirical design. Section 3 presents the main evidence. Section 4 discusses robustness and economic interpretation. Section 5 draws portfolio implications. Section 6 translates the coefficients into economic magnitudes, evaluates a pseudo-real-time conditional-duration rule, and documents policy-regime dependence across FOMC cycle windows. Section 7 states the main limitations. Section 8 concludes.

2. Data and Empirical Design

The empirical strategy proceeds in four steps. The paper first defines the sample and the stock and bond returns used to measure the hedge. It then constructs three observable rate-based proxies that map into real-rate, inflation, and short-rate news. Next, it estimates a pooled pre/post interaction regression that asks whether equity exposures to those proxies changed after 2021. Finally, it uses a time-varying state-space model to summarize how the correlation shift is distributed across channels.

2.1. Sample and assets

The sample runs from 2 January 2003 through 31 December 2025 at the daily frequency. The raw calendar panel contains 6,000 business days, of which 4,696 fall before 2021 and 1,304 fall in 2021–2025. After first-differencing yield series, applying a 252-day rolling standardization window, and dropping days with missing values induced by merges, the baseline regression sample contains 5,245 observations.

The equity side includes 23 baskets: the Fama–French market excess return (French, n.d.a), four broad US equity ETFs (SPY, RSP, QQQ, and IWM), 12 Fama–French industry portfolios (French, n.d.b), and six Fama–French size–book-to-market portfolios (French, n.d.c). The bond side uses synthetic constant-maturity Treasury excess returns constructed from daily Treasury yields and, in robustness checks, Treasury ETFs with medium and long duration.

For maturity τ , the synthetic bond total return uses a carry–duration–convexity approximation (Carr and Wu, 2023),

$$r_{t+1}^{b,\tau} = -\text{MD}_{\tau,t} \Delta y_{\tau,t+1} + \frac{1}{2} \text{CX}_{\tau,t} (\Delta y_{\tau,t+1})^2 + y_{\tau,t}/252, \quad (1)$$

where MD and CX are the modified duration and convexity of a par bond priced off the current yield. Excess returns are formed by subtracting the daily 1-month Treasury bill return.

2.2. Shock proxies

The design uses three reduced-form daily macro-financial proxies:

$$\begin{aligned} z_t^{g,\text{raw}} &= \Delta y_{10,t}^{\text{real}}, \\ z_t^{\pi,\text{raw}} &= \Delta(y_{10,t}^{\text{nom}} - y_{10,t}^{\text{real}}), \\ z_t^{p,\text{raw}} &= \Delta y_{2,t}^{\text{nom}}. \end{aligned} \tag{2}$$

The first series is the change in the 10-year real yield; the second is the change in 10-year breakeven inflation; and the third is the change in the 2-year nominal yield. These proxies provide a transparent way to separate real-rate, inflation, and short-rate news in observable market data. Real-yield changes often capture growth-sensitive discount-rate news and the expected medium-term policy stance; breakeven changes capture inflation-sensitive news; and 2-year nominal yield changes are heavily exposed to near-term policy-path revisions (Hanson and Stein, 2015; Nakamura and Steins-son, 2018; Pflueger and Rinaldi, 2022).

Each raw proxy is winsorized at the 0.5% and 99.5% full-sample quantiles and then standardized with a 252-day past-only rolling z-score. That scaling lets the coefficients be read as responses to a one-standard-deviation move in the respective proxy.

2.3. Pooled interaction regression

The main test estimates whether equity responses to the three proxies changed after 2021. For basket k ,

$$r_{k,t+1}^s = \alpha_k + \sum_{j \in \{g,\pi,p\}} \beta_{k,j} z_t^j + \sum_{j \in \{g,\pi,p\}} \delta_{k,j} (\text{Post}_t \times z_t^j) + \gamma_k \text{Post}_t + u_{k,t+1}, \tag{3}$$

where Post_t equals one from 1 January 2021 onward. The coefficients $\beta_{k,j}$ describe pre-2021 exposures; the interaction terms $\delta_{k,j}$ measure how those exposures changed after 2020. Newey–West standard errors with six lags are reported throughout (Newey and West, 1987).

The key coefficient is $\delta_{k,g}$. If the hedge breakdown reflects a regime in which real-rate/growth-related news became worse for equities, then $\delta_{k,g}$ should be negative and economically meaningful.

2.4. Time-varying decomposition

To complement the pooled regression, the paper estimates a bivariate state-space model for stock and bond excess returns,

$$\begin{pmatrix} r_{t+1}^s \\ r_{t+1}^b \end{pmatrix} = B_t z_t + u_{t+1}, \quad B_{t+1} = B_t + \eta_{t+1}, \tag{4}$$

where $z_t = (z_t^g, z_t^\pi, z_t^p)'$ and the elements of B_t evolve as random walks. The model is estimated by maximum likelihood and smoothed with the Rauch–Tung–Striebel algorithm (Sarkka, 2013,

Chapter 8).

For channel j , the daily own-channel covariance contribution is

$$c_{j,t} = \beta_{j,t}^s \beta_{j,t}^b (z_t^j)^2. \quad (5)$$

These contributions are aggregated to the monthly frequency and normalized by realized stock and bond volatility to form correlation-scale channel contributions. The decomposition then compares average monthly contributions before and after 2021. Because the proxies are not orthogonalized, the resulting shares are descriptive rather than exact structural attribution.

3. Main Evidence

The results are organized in four steps. The paper first documents the regime break in the stock–bond correlation. It then shows that the largest change appears in the equity response to the real-rate proxy and that the pattern is broad across the equity cross section. Next, it uses the time-varying decomposition to compare channel contributions. It closes with subperiod evidence that helps place the change in time.

3.1. The post-2021 regime shift

Exhibit 1 plots the rolling stock–bond correlation. This figure comes first because the first question is whether there is a genuine regime break to explain. In the monthly realized-correlation series, the mean is -0.344 in 2003–2020 and $+0.033$ in 2021–2025, with a difference of $+0.377$ and a simple difference-in-means t -statistic of $+7.49$. That pattern is less consistent with the idea that the 2022 experience was just temporary noise around a stable negative-correlation regime.

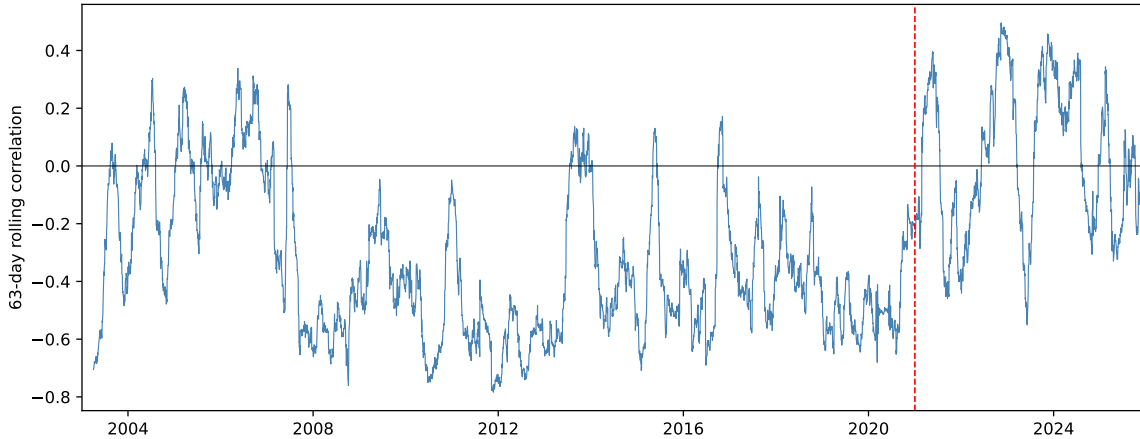


Exhibit 1. Rolling stock–bond correlation. The figure plots the 63-day rolling correlation between the market excess return and the synthetic 10-year Treasury excess return. The dashed vertical line marks 1 January 2021. The reported regime means in the text use monthly realized correlations rather than the smoothed daily series.

3.2. Pooled interaction results

Exhibit 2 reports the pooled interaction regression for the market portfolio and four broad US equity ETFs. The pattern is consistent across all five baskets. The post-2021 interaction on the real-rate proxy is negative in every case and economically large. For the market portfolio, the pre-2021 loading on the real-rate proxy is +0.00185, the post-2021 change is -0.00354 , and the implied post-2021 loading is -0.00170 . The inflation interaction is also negative, but smaller in magnitude. The short-rate interaction is near zero.

Basket	β_g pre	δ_g	$t(\delta_g)$	δ_π	$t(\delta_\pi)$	δ_p
MKT	+0.00185	-0.00354	-5.18	-0.00152	-2.65	-0.00035
SPY	+0.00167	-0.00335	-4.95	-0.00150	-2.64	-0.00036
RSP	+0.00200	-0.00382	-5.45	-0.00211	-3.77	-0.00035
QQQ	+0.00172	-0.00379	-4.96	-0.00123	-1.82	-0.00047
IWM	+0.00227	-0.00436	-4.90	-0.00180	-2.60	-0.00037

Notes: β_g is the pre-2021 loading on the real-rate proxy. δ_g , δ_π , and δ_p are post-2021 interaction terms from Equation (3). Newey–West t -statistics use six lags. Coefficients are in per-one-standard-deviation units because the shock proxies are rolling z-scores.

Exhibit 2. Pooled interaction regression for the market and four broad equity indices.

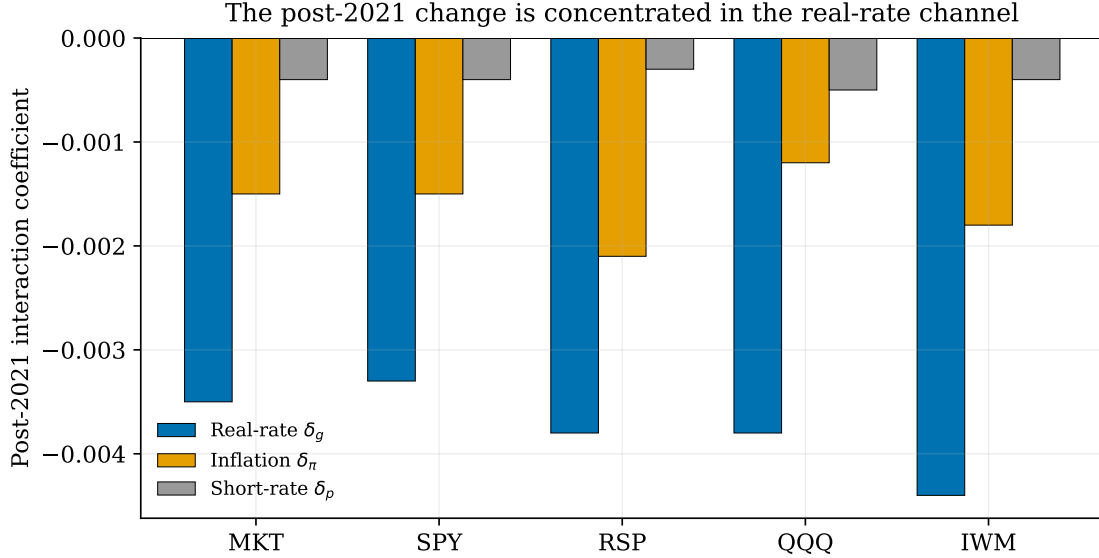


Exhibit 3. Post-2021 interaction coefficients by channel. For each of the five headline baskets, the figure plots the three post-2021 interaction terms from Equation (3): the real-rate term δ_g , the inflation term δ_π , and the short-rate term δ_p . The real-rate interaction is the most negative term for every basket.

Figure 3 shows the same pattern across the three channels. The real-rate interaction is the most

negative term in every basket, the inflation interaction is smaller, and the short-rate interaction is near zero. The pooled results point to a narrow conclusion. The post-2021 increase in stock–bond comovement is associated primarily with a change in the equity response to the real-rate proxy. That result does *not* mean that bond-side exposures were constant, nor does it mean that inflation ceased to matter. It means that, in the empirical design used here, the largest change is concentrated in the real-rate/growth-related channel.

3.3. Breadth across equity baskets

The result is not confined to one index. Across all 23 equity baskets, every single basket has a negative post-2021 interaction on the real-rate proxy and a two-sided HAC p -value below 0.10. That includes every one of the 12 industry portfolios and every one of the six size–book-to-market portfolios. Exhibit 4 visualizes the distribution.

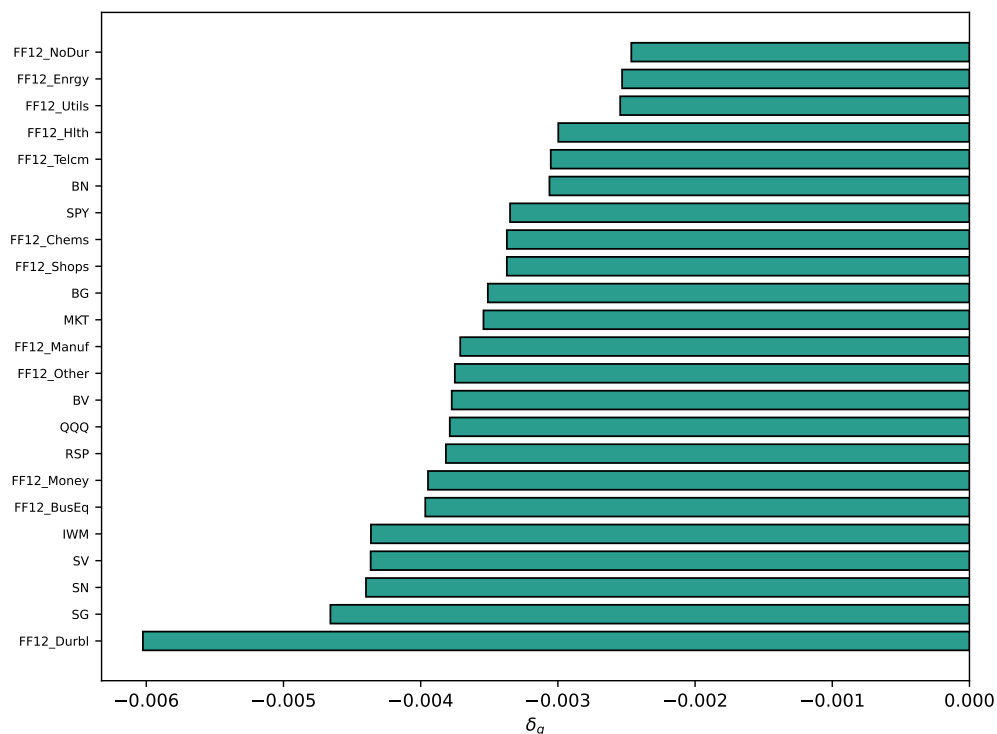


Exhibit 4. Cross-basket evidence on the real-rate interaction. The figure reports the estimated δ_g from Equation (3) for all 23 equity baskets. All estimates are negative; all have two-sided HAC p -values below 0.10.

The breadth of the result helps distinguish a market-wide repricing story from a narrow composition story. If only one segment of the market had changed sign, the result could be dismissed as a composition story, for example a duration-heavy growth-stock effect. Instead, the sign reversal appears in small caps, equal-weighted equities, defensives, cyclicals, value, growth, and industry portfolios alike. That pattern fits a market-wide repricing of discount-rate news better than a

narrow sector story.

3.4. Time-varying decomposition

Exhibit 5 reports the time-varying channel decomposition. The real-rate/growth-related channel explains 0.3359 of the 0.3774 total correlation shift, or about 78.4% of the decomposition total. The inflation channel explains 0.0886, or about 20.7%. The short-rate channel contributes only 0.0037, or roughly 0.9%. The decomposition totals sum to 0.4283, yielding a ShiftFit of 1.135. In other words, the three own-channel terms slightly overshoot the observed correlation shift but remain close enough to be informative, especially given omitted cross-channel covariance terms.

Channel	Pre mean	Post mean	Change (Δ)	Share
Inflation proxy	−0.1712	−0.0825	+0.0886	20.7%
Real-rate proxy	−0.1379	+0.1980	+0.3359	78.4%
Short-rate proxy	−0.0013	+0.0024	+0.0037	0.9%
Total decomposition			+0.4283	100.0%
Observed correlation shift			+0.3774	
ShiftFit			1.1349	

Notes: Monthly channel contributions are the volatility-normalized own-channel covariance components from the time-varying state-space model. Shares are computed relative to the sum of the three channel changes, not relative to the observed correlation shift.

Exhibit 5. Time-varying decomposition of the post-2021 correlation shift.

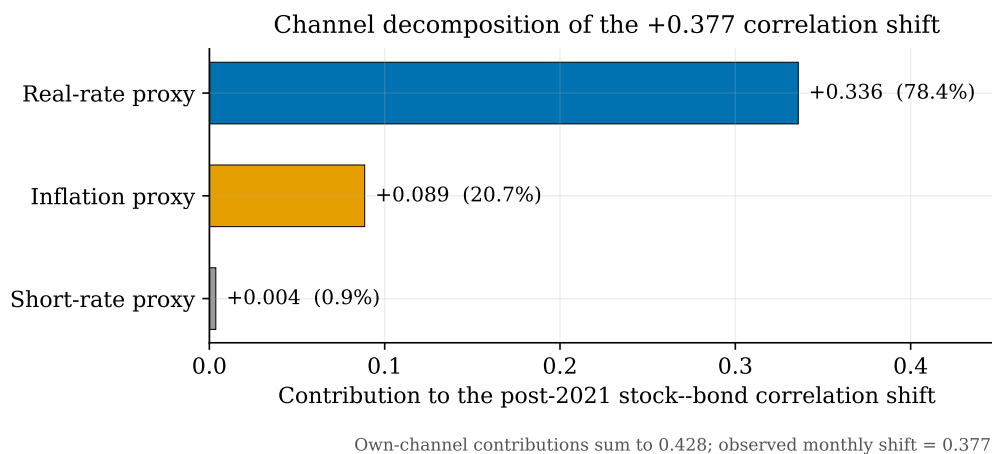


Exhibit 6. Channel decomposition of the correlation shift. The bars show each channel’s own-channel contribution to the post-2021 change in the stock–bond correlation: the real-rate channel accounts for about 78%, the inflation channel for about 21%, and the short-rate channel for about 1%. The three own-channel terms sum to 0.428, slightly above the observed monthly shift of 0.377 because cross-channel covariance terms are omitted.

Figure 6 shows the same decomposition graphically. Exhibit 7 shows the smoothed market loading on the real-rate proxy. The figure shows that the change is not a one-month accident. The loading moves lower through 2022 and remains clearly negative in the later post-2021 period.

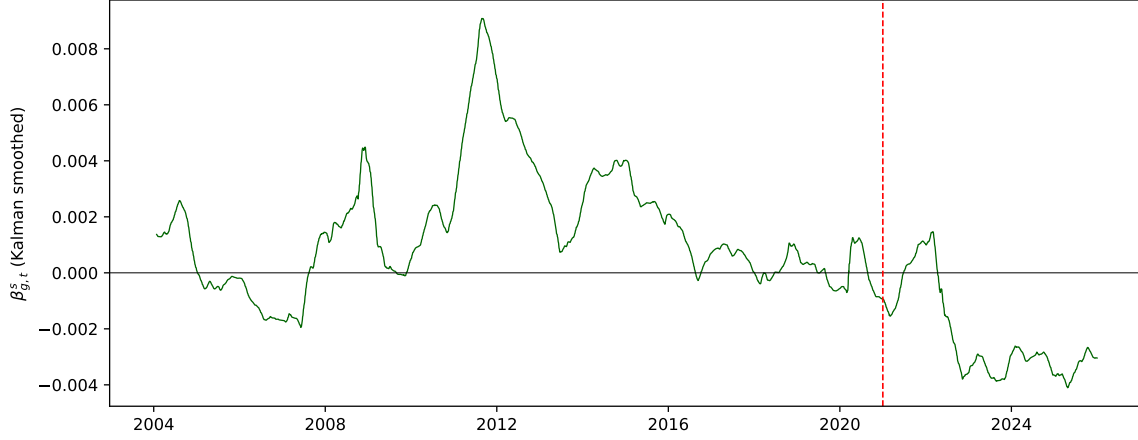


Exhibit 7. Time-varying market loading on the real-rate proxy. The figure plots the Kalman-smoothed market loading associated with the real-rate proxy.

3.5. Subperiod dynamics

The timing of the change is also informative. Exhibit 8 re-estimates the market loading on the real-rate proxy in three post-2020 windows. The loading is statistically indistinguishable from zero in 2021, more negative in 2022 but still imprecise, and strongly negative in 2023–2025.

Window	β_g	t	p	N
2021	−0.00013	−0.16	0.870	240
2022	−0.00153	−1.10	0.271	238
2023–2025	−0.00415	−4.29	< 0.001	713

Notes: Estimates are from separate regressions within each post-2020 subperiod. β_g is the market loading on the real-rate proxy. Standard errors are Newey–West with six lags.

Exhibit 8. Subperiod estimates for the market loading on the real-rate proxy.

This timing helps interpret the results. If the entire post-2021 shift had appeared immediately in early 2021, a pure inflation-salience explanation would be more natural. Instead, the real-rate/growth-related loading becomes decisively negative only after the tightening cycle is underway.

4. Robustness and Economic Interpretation

4.1. What the robustness checks show

A straightforward challenge is that higher nominal rates after 2021 may have mechanically raised equity sensitivity to rate news, independently of any change in the policy reaction function. Three checks address that possibility.

First, the specification adds an interaction between the post-2021 indicator and the level of the 10-year nominal Treasury yield. If the headline result were merely a mechanical consequence of operating in a higher-rate world, that control should absorb the change. It does not: the real-rate interaction remains essentially unchanged at -0.00354 with $t = -5.16$. The added level term is economically tiny.

Second, the analysis excludes the top decile of days ranked by the absolute value of the short-rate proxy, treating them as the days most likely to be dominated by policy-announcement information. Again the result survives almost unchanged: the estimated real-rate interaction remains strongly negative at -0.00347 with $t = -4.62$.

Third, the analysis isolates the pre-liftoff window from January 2021 through February 2022. In that window, the estimated market loading on the real-rate proxy is $+0.00022$ with $t = +0.29$, so the sign flip is absent before policy tightening begins. This is more informative about timing than about identification. It is not a decisive rate-level test because the window is short and remains a low-rate period. Even so, the timing is informative: the shift does not appear to predate the tightening cycle.

Taken together, the robustness results do not prove a specific structural model, but they do weaken the simplest alternatives. The evidence is hard to reconcile with a story in which the post-2021 change is only a mechanical consequence of higher rate levels or only a by-product of a handful of policy-announcement days.

4.2. Economic interpretation

The most natural reading is a change in how markets map macro news into expected policy and discount rates. In earlier years, stronger growth-related news could support equities because the cash-flow effect dominated. In the post-2021 tightening regime, stronger growth-related news increasingly signaled higher real discount rates and a more persistent hawkish stance from the Federal Reserve. In that environment, news that is positive for the economy can still be negative for equities because it raises the expected path of policy and real yields (Bernanke and Kuttner, 2005; Boyd, Hu, and Jagannathan, 2005; Pflueger and Rinaldi, 2022).

The timing strengthens that interpretation. The loading on the real-rate proxy is not meaningfully negative before liftoff, becomes more negative during 2022, and is clearly negative only in 2023–2025. In plain language, after inflation returned, the equity market increasingly treated stronger growth as news that could prolong tight policy rather than as unambiguously good cash-flow news.

This interpretation fits the broader stock–bond literature. Campbell, Sunderam, and Viceira show that nominal bonds change character as inflation and policy regimes change, while Campbell, Pflueger, and Viceira emphasize the joint role of inflation and the output gap in determining whether bonds act as safe assets or risky assets (Campbell, Sunderam, and Viceira, 2017; Campbell, Pflueger, and Viceira, 2020). The evidence here adds a narrower point: in the recent US episode, the dominant observable shift is not just that inflation news mattered more, but that real-rate/growth-sensitive news became materially worse for equities.

Because the shocks are reduced-form proxies, omitted state variables could still matter, and the post-2021 window is still one episode. The evidence is consistent with a hawkish-reaction-function story, but it does not isolate that mechanism on its own.

5. Portfolio Implications

The results have three implications for asset allocators.

First, long-duration Treasuries should not be treated as unconditional hedges. In the post-2021 sample, the average stock–bond correlation is close to zero, not deeply negative. That is enough to reduce the protection value of duration materially, especially in fast drawdowns driven by policy repricing.

Second, investors should monitor the *state dependence* of the hedge rather than only the trailing correlation itself. When markets believe the central bank will react forcefully to resilient growth, real-rate news is more likely to hurt both bonds and equities at the same time. In that regime, portfolio resilience depends less on static duration exposure and more on whether the rate path is likely to amplify or offset equity risk.

Third, internal equity rotation is unlikely to solve the problem by itself. Because the sign reversal appears across all 23 baskets in the sample, moving from growth to value, from cap-weighted to equal-weighted indices, or from one industry group to another does not eliminate the underlying comovement channel. Investors who need more reliable protection may need to pair duration with instruments or strategies that do not load on the same real-rate repricing mechanism.

The paper argues that the hedge is conditional, and that the relevant conditioning variable is not only inflation but also the market’s perception of the policy reaction function.

6. Economic Magnitudes, Conditional Hedging, and Policy Regime

The preceding sections document a precise statistical pattern: the post-2021 loading on the real-rate proxy is negative and the change from the pre-period is large. This section translates that pattern into three practitioner-relevant statements. Section 6.1 converts the coefficients to basis points and hedge-ratio units. Section 6.2 shows that a simple, pseudo-real-time conditional-duration rule preserves Sharpe while cutting downside in the post-2021 sample. Section 6.3 uses Federal Open Market Committee (FOMC) cycle windows to show that the loading on the real-rate proxy varies with the stance of monetary policy, moving from strongly positive in the pre-liftoff/QE era to

strongly negative in the post-2022 hold-and-cut phase.

6.1. Translating coefficients to basis points

The interaction coefficients in Equation (3) are in per-one-standard-deviation units. In the baseline sample, the full-sample winsorized standard deviation of the daily change in the ten-year real yield is about 4.88 basis points. A typical 10-basis-point real-yield move is therefore approximately 2.05 standard deviations of z_t^g .

Using that conversion, the pre-2021 daily equity response to a 10-basis-point real-yield move is approximately +38 basis points, and the post-2021 response is approximately −35 basis points, for a total change of −73 basis points. The same conversion translates the shift from a stock–bond daily correlation of −0.344 to +0.033 into a change in the hedge ratio (the slope of market returns on bond returns) from approximately −0.97 pre-2021 to +0.11 post-2021. A naive 60/40 portfolio delivered roughly 44% risk reduction relative to 100% equity in the pre-2021 sample; in the post-2021 sample, that risk reduction falls to about 36%. Exhibit 9 summarizes the conversion.

Quantity	Pre-2021	Post-2021	Change
Equity response to 10 bp real-yield move	+37.8 bp	−34.7 bp	−72.6 bp
Stock–bond daily correlation	−0.344	+0.033	+0.377
Hedge ratio $\beta_{\text{MKT on bond}}$	−0.97	+0.11	+1.08
60/40 annualized volatility	10.7%	11.3%	+0.6 pp
Risk reduction of 60/40 vs. 100% equity	44.1%	36.3%	−7.7 pp

Notes: Equity response is computed as $\hat{\beta}_g \cdot (10/\hat{\sigma}_{\Delta y_{10}^{\text{real}}})$ where $\hat{\sigma}$ is the winsorized full-sample standard deviation of the daily change in the ten-year real yield, and the resulting decimal daily return is multiplied by 10,000 to report in basis points. Hedge ratio is the OLS slope of daily market excess returns on daily synthetic ten-year Treasury excess returns within each sample.

Exhibit 9. Economic magnitude of the pre/post shift.

The numbers in Exhibit 9 are descriptive rather than structural, but they address the question that motivates the paper in practical terms: for a dollar of duration exposure, the pre-2021 world delivered meaningful diversification against growth-sensitive rate news, and the post-2021 world does not.

6.2. A pseudo-real-time conditional-duration rule

If the hedge is regime-dependent, a natural next question is whether a simple rule that recognizes the regime can preserve portfolio returns while reducing the downside. This subsection reports a minimal, pseudo-real-time backtest to address that question. It is deliberately simple: one signal, one threshold, two portfolio weights, no optimization. We use “pseudo-real-time” because the Kalman state at day t is a filtered (past-only) estimate, but the Kalman hyperparameters are calibrated once on the full sample (see the “Rule” paragraph below).

Rule. Let $\tilde{\beta}_{s,t-1}^g$ denote the 63-day rolling mean of the Kalman-filtered market loading on the real-rate proxy, evaluated using only information available through day $t - 1$. Because the Kalman filter state update is a past-only operator and the rolling mean is further lagged by one day, the signal applied on date t uses no information beyond the close of day $t - 1$. We note one caveat: the Kalman hyperparameters (the state-noise covariance Q and observation-noise covariance H) are estimated once by maximum likelihood on the full sample. A fully recursive implementation would re-estimate them through time; the exercise reported here is therefore pseudo-real-time rather than strict real-time at the parameter-estimation level. The conditional portfolio holds 60% equity at all times and switches bond weight as follows:

$$w_t^b = \begin{cases} 20\% & \text{if } \tilde{\beta}_{s,t-1}^g < -0.0015, \\ 40\% & \text{otherwise,} \end{cases}$$

with any residual held in one-month Treasury bills. The static benchmark is a constant 60/40 portfolio. Both portfolios are compared against a 100% equity benchmark. The backtest starts on 3 January 2005 to allow a 252-day Kalman burn-in, yielding 5,020 aligned daily observations through 31 December 2025; after dropping the 63-day signal warm-up, 4,957 effective trading days are used for portfolio evaluation. The threshold of -0.0015 is a full-sample illustrative calibration chosen so that the rule is active on approximately 2% of pre-2021 effective days and 67% of post-2021 effective days, matching the post-2021 concentration of persistently negative filtered betas; we report it as an illustrative calibration rather than as a pre-specified trading rule.

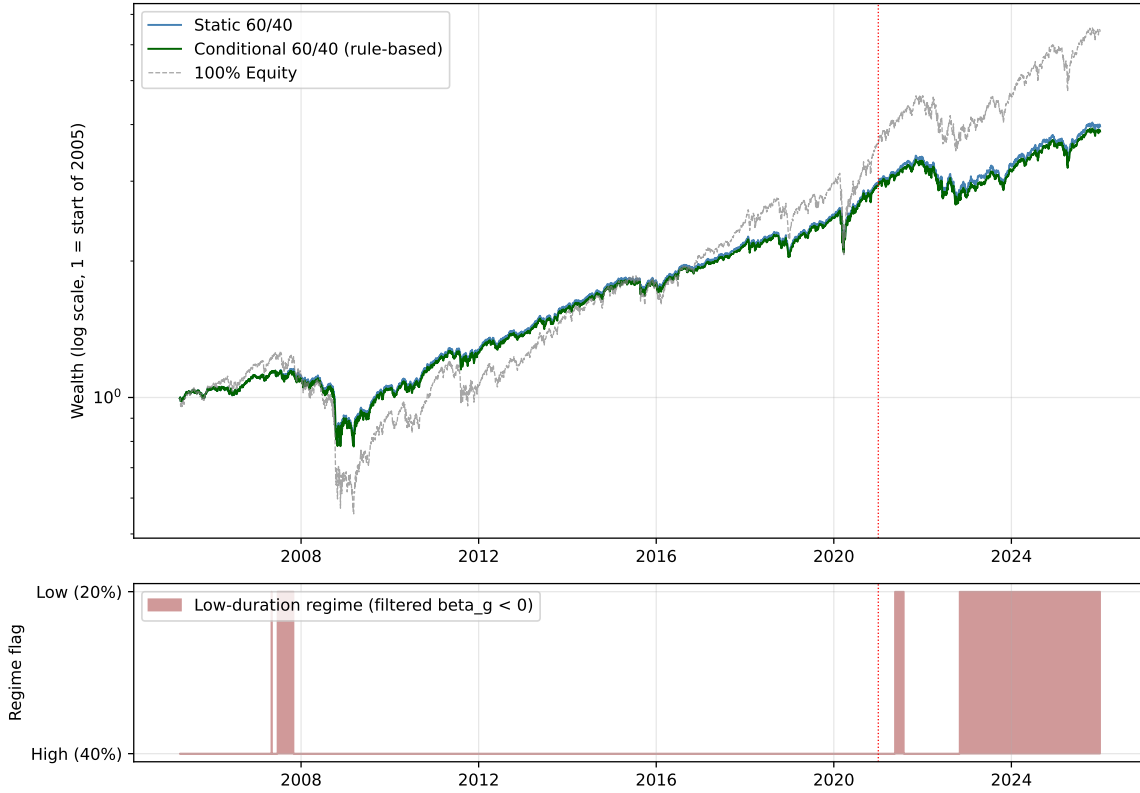


Exhibit 10. Wealth paths for static 60/40, the conditional-duration rule, and 100% equity. Top panel plots cumulative wealth on a log scale starting from unity in January 2005. The dashed vertical line marks 1 January 2021. Bottom panel marks the days on which the conditional rule holds the lower 20% bond weight.

Results. Exhibit 11 summarizes the outcome. Over the full 2005–2025 sample, the conditional rule produces essentially the same annualized return as static 60/40 and a marginally lower volatility, so Sharpe ratios are indistinguishable. The relevant comparison, however, is the post-2021 sub-sample, in which the rule fires frequently. In the post-2021 window, the conditional rule matches the static portfolio’s return while delivering a modestly lower realized volatility and a smaller downside deviation. In the 2023–2025 sub-sample, where the filtered negative regime is most persistent, the conditional rule reduces annualized volatility from 9.84% to 9.51%, reduces downside deviation from 6.57% to 6.37%, and reduces the worst quarterly drawdown from -4.13% to -3.64% . The rule is not designed to generate excess return; it is designed to preserve the risk-reduction function of the bond allocation when real-rate news becomes bad news for equities.

Period	Portfolio	Ann. ret	Ann. vol	Sharpe	Max DD	Worst Qtr
Full sample	Static 60/40	+7.25%	11.11%	0.65	−31.7%	−12.5%
	Conditional 60/40	+7.11%	11.06%	0.64	−32.0%	−12.5%
	100% Equity	+9.90%	19.38%	0.51	−56.1%	−25.3%
Pre-2021	Static 60/40	+5.71%	11.04%	0.52	−31.7%	−9.7%
	Conditional 60/40	+5.63%	11.05%	0.51	−32.0%	−9.7%
Post-2021	Static 60/40	+1.46%	11.33%	0.13	−20.4%	−12.5%
	Conditional 60/40	+1.40%	11.09%	0.13	−20.4%	−12.5%
2023–2025	Static 60/40	+1.67%	9.84%	0.17	−13.3%	−4.1%
	Conditional 60/40	+1.67%	9.51%	0.18	−13.2%	−3.6%

Notes: Annualized return, volatility, and Sharpe are computed on daily excess returns over the one-month Treasury bill. The conditional rule uses a 63-day rolling mean of the Kalman-filtered market loading on the real-rate proxy, with the rolling mean lagged by one day. When that signal is below -0.0015 , the bond weight is reduced from 40% to 20% with the difference held in cash.

Exhibit 11. Backtest of static vs. conditional 60/40, 2005–2025.

The simple rule is meant as an illustration rather than a recommended allocation. Several features matter for interpretation. First, the rule is deliberately coarse: one filtered signal, one threshold, and a binary duration switch. It does not adjust for level of rates, term premium, or equity volatility. Second, the signal is slow: the 63-day rolling mean filters short-term noise in the filtered beta, which helps avoid whipsaw but also delays regime detection. Third, the bond weight is reduced into cash rather than into alternative hedges; under a different menu of available instruments the improvement could be larger. The point is narrow: even a coarse pseudo-real-time rule preserves Sharpe and reduces downside once the regime has shifted, which indicates that the regime shift is operationally recognizable in real time.

6.3. Policy-regime dependence: FOMC cycle windows

The third piece of evidence addresses interpretation rather than allocation. If the post-2021 sign change in the equity response to real-rate news is best understood through a shift in the perceived Federal Reserve reaction function, the response should vary with the policy stance itself. Exhibit 12 estimates the market loading on the real-rate proxy within five non-overlapping windows defined by FOMC stance rather than by calendar year.

FOMC regime window	Dates	N	$\hat{\beta}_g$	t
Pre-liftoff era (incl. ZLB/QE)	2005-01 – 2015-11	2,613	+0.00259	+5.51
Normalization	2015-12 – 2019-07	877	+0.00024	+0.47
Easing + COVID ZLB	2019-08 – 2022-02	616	+0.00133	+1.60
Hiking cycle	2022-03 – 2023-07	339	−0.00238	−1.91
Hold + Cuts	2023-08 – 2025-12	575	−0.00412	−3.51

Notes: Windows are defined by federal funds stance and the direction of the FOMC policy path. The pre-liftoff era (2005-01 to 2015-11) contains the 2005–2007 pre-crisis low-rate period, the 2008-12 transition to the 0–0.25% target range, and the subsequent ZLB/QE phase through the December 2015 liftoff. Normalization runs from liftoff (December 2015) through the first 2019 rate cut (31 July 2019). The easing + COVID ZLB window (August 2019 through February 2022) contains the 2019 easing, the March 2020 return to the 0–0.25% target range, and the subsequent ZLB period with large-scale asset purchases; it ends the day before the first hike of the 2022 cycle (16 March 2022). The hiking cycle runs through the last hike (26 July 2023). Hold + Cuts covers the subsequent plateau and the first phase of cuts (the initial cut was in September 2024). Each row reports the within-window OLS estimate of Equation (3) without interaction terms, using z_t^q , z_t^π , and z_t^p jointly with Newey–West standard errors (six lags).

Exhibit 12. Market loading on the real-rate proxy within FOMC regime windows.

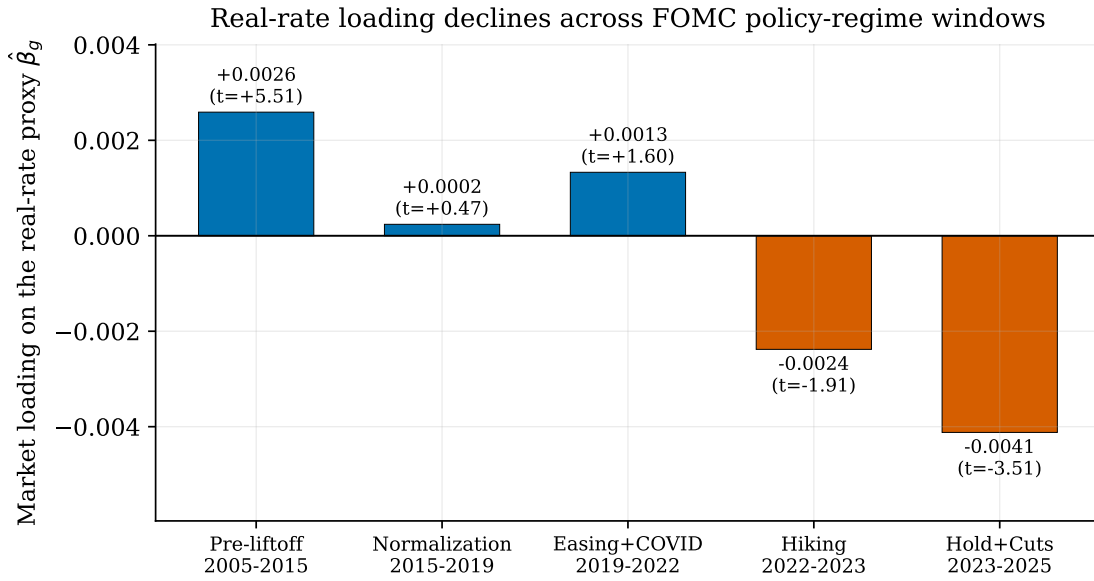


Exhibit 13. Real-rate loading across FOMC policy-regime windows. The bars plot the within-window market loading on the real-rate proxy $\hat{\beta}_g$ for the five policy-regime windows, with t -statistics shown above each bar. The loading falls from strongly positive in the pre-liftoff era to strongly negative in the post-2022 hold-and-cut phase.

Figure 13 plots these estimates. The pattern is broadly declining from strongly positive in the pre-liftoff era to strongly negative in the post-2022 hold-and-cut phase, with one non-monotonic ex-

ception. In the pre-liftoff era of 2005–2015, the loading is strongly positive. This window aggregates a pre-crisis low-rate period (2005–2007) and the post-2008 ZLB/QE phase; in both sub-periods the market could interpret stronger growth-sensitive rate news as primarily cash-flow news rather than as a signal of tighter policy. During the 2015–2019 normalization window, the loading falls close to zero. It rises modestly to +0.00133 during the easing + COVID ZLB window of 2019–2022, the one window that breaks strict monotonicity; the rise is consistent with a return to a cash-flow-dominated interpretation when policy is easing or pinned at the effective lower bound. Once the hiking cycle begins in March 2022, the loading turns negative. In the 2023–2025 hold-and-cut window, the loading is -0.00412 with a t -statistic of -3.51 , nearly twice as negative as the hiking-window estimate.

This pattern is consistent with the hawkish-reaction-function interpretation of the post-2021 shift. When the policy rate is at or near the effective lower bound, stronger real-yield news does not translate into higher expected policy rates; the effective channel is cash flows and the loading is positive. Once the Federal Reserve is both able and willing to respond to real activity, stronger growth-sensitive rate news is increasingly read as a signal of more restrictive policy, and the effective channel flips from cash flows to discount rates. The broad pattern across the five windows is not a simple calendar effect, since the windows are uneven and include two distinct ZLB-adjacent periods (pre-liftoff and easing + COVID ZLB) and a tightening phase. It is a policy-regime effect.

A complementary split by the magnitude of the short-rate proxy $|z_t^p|$ is less straightforward. The sign change is stronger on low-policy-news days than on high-policy-news days in the post-2021 sample, and a triple-interaction regression ($\text{Post}_t \times \text{HighNews}_t \times z_t^q$) yields a coefficient of +0.00257 with $t = +2.09$. One reading is that high-policy-news days carry confounded signals: both cash-flow and discount-rate components move at once, so the isolated growth channel is harder to see. Another reading is that the mechanism does not operate primarily on announcement days. The FOMC-window evidence in Exhibit 12 speaks more cleanly to policy-regime dependence than the policy-news split does, and the policy-news split is reported here as a robustness check rather than as primary identification.

Taken together, the three subsections convert a statistical finding into a practitioner-oriented statement, demonstrate that a naive pseudo-real-time rule preserves Sharpe while cutting downside once the regime shifts, and document broadly declining regime dependence across FOMC cycle windows. None of these results substitutes for a structural identification of the monetary-policy reaction function, which lies outside the scope of this paper. They do, however, indicate that the post-2021 pattern is economically sizeable, operationally recognizable, and concentrated in the regimes one would expect under a more hawkish reaction function.

7. Limitations

The main limitations are three. First, the macro proxies are reduced form. A change in the 10-year real yield is not a pure growth shock, and a change in breakeven inflation is not a pure inflation

surprise. Second, the decomposition is partial. Because the three proxies are not orthogonalized, the channel shares omit cross-channel covariance terms. Third, the evidence is entirely US based and concentrated in one post-2021 episode, so it does not by itself show whether the same mechanism dominates in other countries or reverses under a different policy regime.

8. Conclusion

This paper studies why long-duration Treasuries stopped hedging US equities after 2021. In the data used here, the stock–bond correlation moves from -0.344 before 2021 to $+0.033$ after 2020. The results point less to a pure inflation-only story than to a sign change in the market’s response to growth-sensitive real-rate news. The pre-2021 loading on the real-rate proxy is positive, the post-2021 loading is negative, and the difference is large and precisely estimated. The inflation proxy also matters, but less. The short-rate proxy contributes little on its own.

The paper does not claim to identify a pure structural growth shock. The narrower claim is that the post-2021 hedge breakdown is more closely associated with real-rate/growth-sensitive discount-rate news than with a pure inflation-only story, and that this association becomes strongest once the tightening cycle is underway.

For investors, the diversification value of long Treasuries is regime dependent. When strong macro news is likely to be interpreted through a hawkish policy lens, duration is a weaker hedge than many allocators were used to during the pre-2021 period.

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